Current Verdicts and Legal-Intervention Sufficiency in Battery-Free Design

Development manuscript

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Abstract

We study when an engineering representation supports a current service verdict and when it supports a legal design change. This public development manuscript binds all reported numbers to executable evidence. Agent benefit, format benefit, physical qualification, and autonomous observation discovery are separate untested claims.

1 Problem

A fixed task T and world w define a service verdict $V(w, d, T)$. A representation $R(w, d)$ is current-verdict sufficient on a declared domain if equality of representations implies equality of current verdicts. For a public legal intervention set $\mathcal{D}$, intervention sufficiency requires the same implication for every common legal change $\delta \in \mathcal{D}$ applied to both designs. The second property need not follow from the first.

2 Observation acquisition

Observations add facts at measured acquisition and qualification cost. We distinguish selection of developer-defined observations from constructing executable observations. A static sufficient statistic or a complete executable model may provide all necessary information.

3 Implementation

The frozen protocol specifies ideal clamped energy and reset-state models, legal D changes, independent implementations, and negative controls. It changes no existing physical or task semantics.

4 E2–E5: public development results

4.1 E2–E3: two unexcited mechanisms

Run `IF-7412532c4a6ab7a5879e` contains two public development pairs, not statistical replications or unseen tests. Each pair has identical current observed event/rail traces and two passing current tasks. The identical legal D change then yields FAIL/PASS.

For the clamp pair, a fixed 20 μW source supplies 40 μJ before a 30 μJ service transaction at 2.5 s, with a 1 V service reserve. Both systems use 10 μF initially. Their ideal clamps are 3 V and 5 V. The initial maximum rail voltage is $\sqrt{8} = 2.828$ V, so neither clamp activates and the entire current trajectory coincides. Changing only capacitance to 4 μF is legal in the common $[4, 12]$ μF domain. The 3 V clamp now spills 22 μJ and makes only 16 μJ available above reserve; it fails the fixed service. The 5 V clamp spills zero and delivers 30 μJ; it passes. The post-change margins are -14 and 8 μJ. These are ideal energy transactions, not measured PMIC startup or MCU current.

For the reset pair, an event is captured at 1 s and must be transmitted once at 4 s, by a 5 s deadline. The external event and transmission time stay fixed. Without a power reset, volatile and retained pending

memories both deliver the event. A common D change inserts an outage from 2 to 3 s: volatile pending memory is cleared and delivers zero events; retained pending memory delivers one. This changes a state transition rather than rescaling an old energy/window template.

The analytic clamp implementation agrees with projected time integration; the ordered memory machine agrees with a separate event-program parser and tick interpreter. Both 1 ms and 0.5 ms settings agree on all eight current/intervention cases. Maximum energy discrepancy is 2.84×10^{-12} μJ. These implementations share ideal assumptions, so agreement is numerical/software evidence only.

Deleting the unexcited clamp threshold or reset-retention information restores aliasing in this encoding. It does not establish universal necessity of a field name or complete device-level E3 validation. Indeed an enriched *static* model is sufficient on the declared ideal domain. It passes all 28 exposed legal-grid checks (20 clamp, eight memory), and a purely additive static control agrees with independent packet accounting on five service-cost values, including passing and failing designs. The clamp statistic is

$$M(C) = \min(E_0 + P_{\text{in}} t_{\text{in}}, CV_{\max}^2/2) - CV_{\min}^2/2 - E_{\text{service}}.$$

Its sufficiency follows here from all input preceding service, an ideal cap, and no leakage or intervening workload. Those assumptions bound the claim.

4.2 E4: executable equal-information preparation

At each information level the prose and JSON packets contain identical facts and an identical summary (eight parity checks across 16 packet files). In fixed-information use no additional tool or read is legal. In active-observation use all arms start with the same current packet and share the same legal observations. Table 1 reports deterministic parse-and-judge checks on the proposed intervention. Abstentions are not counted as errors. This constrained prose serialization is preparation, not a test of model comprehension or unconstrained writing.

Information	Format	Correct	Unknown
current	free-text	0/4	4/4
current	structured	0/4	4/4
sufficient	free-text	4/4	0/4
sufficient	structured	4/4	0/4

Table 1: Fixed-information parser/decision checks; no model calls. Model-format benefit is NOT_RUN.

4.3 E5-1: measured finite observation selection

The selector receives researcher-supplied required-field metadata, not outcomes, and chooses the smallest eligible source by actual file-size estimate. Table 2 totals four public cases. Bytes are payload bytes returned to Python, separately counting source readback for audit; they are not storage-device I/O or monetary fees. Wall time includes acquisition, source audit, and selector where used. Source-export setup is retained in the raw data. These are single host measurements with uncontrolled OS cache, not a latency benchmark.

Observation	Correct	Read B	Audit B	Wall ms	Model runs
uninformative	0/4	4648	4648	0.289	0
mechanism-spec	4/4	103	103	0.260	0
fixed-rich	4/4	4747	4747	0.282	0
full-raw	4/4	8544	8544	0.333	0
executable-dynamic	4/4	8544	8544	1.886	4
selected	4/4	103	103	0.236	0

Table 2: Observation acquisition and audit costs. Uninformative returns UNKNOWN on all four.

Fixed-rich, complete raw information, and an executable dynamic response all judge four of four correctly. The selected observation reads exactly the same facts and bytes as the fixed minimal mechanism-spec baseline. This batch supplies no independent algorithmic selection gain and does not weaken those strong baselines. A zero-byte acquisition budget abstains; three illegal changes and a tampered source binding are rejected.

4.4 E5-2: executable construction preparation

A deterministic compiler turns a researcher-specified headroom expression into a new standalone Python observation, with source hash, input schema and applicability domain. Its values -14 and 8 μJ agree with independent integration on all 20 exposed clamp-grid points and in two fresh-process executions. Construction took 0.239 ms; reference qualification took 13.017 ms (20 model runs); the two fresh processes took 37.730 ms combined. The equation and relevant inputs were manually supplied. This validates executable construction and audit machinery, not autonomous discovery or held-out usefulness.

Evidence binding. The frozen protocol, raw result, packet files, generated program and costs accompany `IF-7412532c4a6ab7a5879e`. Data source: `docs/research/parallel-20261019/interfaces/data/IF-7412532c4a6ab7a5879e/result.json`. Entry point: `scripts/campaign_interfaces.py`; manuscript tables: `papers/interface-sufficiency-20261019/render_results.py`. Paid model requests and native dispatches are zero. Historical successful product solves were not rerun. The old three scaled families and the refinement/fixed-rich tie remain separate development negatives.

5 E4: information and format

Free-text and structured packets share exact facts and summary at each information level. Fixed-information trials prohibit additional reads; active-observation trials share the same initial packet and legal tools. Deterministic parsing is a data integrity check; model-format effects are not measured here.

6 E5: selection and construction

The first stage uses a declared finite observation catalogue. The second compiles a researcher-specified expression into a new executable and checks it against an independent reference. Human-supplied formulas are not Agent discoveries.

7 Boundaries and threats

All examples are exposed public development mechanisms. Two implementations of the same ideal equations check implementation consistency, not device reality. Fixed-rich, full raw information, executable dynamic models, and sufficient static controls remain valid baselines. No novelty priority is claimed without a literature audit. Historical three-family refinement tied the fixed-rich baseline and remains a separate development result.